RADINTEL AI / RADIOLOGY ONLY

Technical Documentation

ASP.NET Core architecture, DICOM/PACS, AI/RAG, data design, APIs and deployment

Version 1.0 | 14 October 2026 | Planning draft - not clinical or legal advice

1. Architecture decisions

Build a .NET modular monolith with clear domain boundaries and asynchronous integration jobs. Avoid premature microservices. Store patient/order/report metadata in PostgreSQL or SQL Server; use an existing PACS/DICOM archive (e.g., Orthanc for development) and OHIF viewer. AI is a separately controlled, optional drafting capability. Deployment supports single-tenant or strongly isolated multi-tenant configurations.

2. Logical architecture

React + TypeScript web UI -> ASP.NET Core API -> Domain modules (Identity, Clinics, Patients, Orders, Scheduling, Worklist, Reports, ImagingIntegration, AI, Notifications, Audit, Analytics) -> relational database. DICOM adapter -> PACS / Orthanc -> OHIF viewer via authorized DICOMweb. Background worker -> integration retries, notifications and audit exports. AI adapter -> transcription / LLM -> validation -> clinician review.

3. Proposed solution structure

RadIntel.sln: src/RadIntel.Api; src/RadIntel.Application; src/RadIntel.Domain; src/RadIntel.Infrastructure; src/RadIntel.Worker; src/RadIntel.Contracts; src/RadIntel.Integrations.Dicom; src/RadIntel.Integrations.Fhir; src/RadIntel.AI; tests/RadIntel.UnitTests; tests/RadIntel.IntegrationTests; tests/RadIntel.SecurityTests; web/radintel-web; deploy/docker; deploy/infra.

4. Module boundaries

Module	Owns	Dependencies
Identity/Tenancy	Organizations, users, roles, scopes	External OIDC provider
Patient Registry	Patient identifiers, dedup workflow	Tenant policies
Orders/Scheduling	Orders, appointments, statuses	Patient Registry
Imaging Integration	Study references, PACS endpoint configs	DICOMweb adapter
Reporting	Templates, versions, signatures, addenda	Orders, authorization
AI Assistant	Transcripts, proposals, model metadata	Reporting, AI provider
Audit/Analytics	Immutable events, aggregate metrics	Domain event outbox

5. Relational schema sketch

Table	Core fields / constraints
Organizations	Id PK, Name, Status, CreatedAt
Facilities	Id PK, OrganizationId FK, Name, TimeZone
Patients	Id PK, OrganizationId FK, ExternalId, NameEncrypted, DOBEncrypted
ImagingOrders	Id PK, OrganizationId FK, PatientId FK, Modality, Status, RequestedAt
Appointments	Id PK, OrderId FK, ScannerId, StartsAt, EndsAt, Status
ImagingStudies	Id PK, OrderId FK, StudyInstanceUID UNIQUE per archive, ArchiveId
Reports	Id PK, OrderId FK, Status, CurrentVersionId
ReportVersions	Id PK, ReportId FK, Version, BodyEncrypted, AuthorId, CreatedAt
Signatures	Id PK, ReportVersionId FK, SignedBy, SignedAt, SignatureHash
AIProposals	Id PK, ReportId FK, ModelVersion, PromptVersion, Status, ReviewerId
AuditEvents	Id PK, OrganizationId, ActorId, Action, EntityId, Timestamp, Hash

Apply OrganizationId filters at data access and authorization layers; verify with cross-tenant penetration tests. Use UUID identifiers, optimistic concurrency tokens, indexes on OrganizationId+Status+CreatedAt and patient timeline. Avoid placing patient names in search logs or tracing.

6. API contract examples

Endpoint	Action	Authorization
POST /api/v1/patients	Create patient	Scheduler/Admin
POST /api/v1/imaging-orders	Create imaging order	Authorized staff
GET /api/v1/worklist	Filter assigned studies	Technician/Radiologist
POST /api/v1/reports/{id}/drafts	Save draft	Radiologist
POST /api/v1/reports/{id}/ai-draft	Request AI proposal	Radiologist + feature flag
POST /api/v1/reports/{id}/sign	Sign immutable version	Radiologist + step-up auth
POST /api/v1/reports/{id}/addenda	Create correction	Authorized radiologist
GET /api/v1/studies/{id}/viewer-session	Issue short-lived viewer session	Authorized clinician

API conventions: OAuth2/OIDC, problem-details errors, idempotency keys on writes, cursor or page-based pagination, ETags/concurrency on mutable records, explicit tenant context, OpenAPI documentation, rate limits and correlation IDs.

7. DICOM/PACS integration

Use DICOMweb QIDO-RS for study search, WADO-RS for retrieval, and STOW-RS where supported and authorized. Do not assume every existing archive supports these APIs. Provide integration adapters for legacy DICOM DIMSE or vendor-specific gateways when required. Use StudyInstanceUID/SeriesInstanceUID and never infer identity solely from a name.

OHIF viewer is embedded or launched with a short-lived authenticated context. The viewer must not receive unrestricted PACS credentials. Ensure appropriate clinical display validation, supported modalities, network bandwidth and browser compatibility before diagnostic use.

8. AI pipeline and guardrails

Pipeline: physician notes or dictated speech -> optional speech-to-text -> normalization -> prompt with approved reporting template -> structured JSON proposal -> schema validation -> UI diff against raw input -> radiologist edits and accepts -> ordinary report draft -> final sign-off. Do not infer medical observations from image pixels in MVP.

RAG: ingest only curated institutional reporting templates, approved terminology and operating policies; split and version documents; generate embeddings; store in access-controlled vector index; retrieve tenant-scoped passages; display citations; reject unsupported answers. Avoid mixing patient data into general knowledge indexes.

Model gateway: configurable provider (Azure OpenAI or approved private model), version pinning, content minimization, no training on customer data by default, per-tenant feature flags, timeout/circuit breaker, prompt-injection defense and safe fallback to manual reporting.

9. Security, privacy and audit

- OIDC/OAuth2, MFA, least-privilege RBAC and resource-level patient authorization.
- Encryption at rest and in transit; managed secrets; key rotation; segregated backups.
- Append-only audit for view, export, change, sign, sharing, failed access and administrative actions.
- Threat model includes cross-tenant leaks, malicious DICOM metadata, prompt injection, insecure links and insider access.
- No production patient data in development/test environments; synthetic/de-identified fixtures only.
- Document lawful processing basis, retention schedule, processor agreements, breach response and clinical safety governance.

10. Reliability and operations

Use outbox pattern for report-signed notifications; background jobs with retries and dead-letter handling; avoid duplicate deliveries through idempotency. Provide health checks, structured logs without PHI, OpenTelemetry traces, metrics for API latency, PACS errors, AI cost and queue depth. Daily encrypted backups and quarterly restoration drills as a starting proposal.

11. Deployment topologies

Topology	Benefits	Tradeoffs
Managed cloud	Rapid rollout, centralized updates	Data residency and network/PACS connectivity
Clinic on-prem	Local control, close to scanners	Maintenance, backups, support burden
Hybrid	Local PACS + hosted workflow/AI gateway	Complex networking and governance

12. Testing strategy

- Unit tests: state machines, authorization policies, report versioning, template validation.
- Integration tests: database, PACS DICOMweb mock, outbox, identity, tenant boundaries.
- Contract tests: FHIR mappings, DICOM identifiers, AI structured-output schemas.
- Security tests: OWASP, cross-tenant access, expiring links, secrets, upload sanitization.
- Clinical acceptance: radiologist reviews report templates, dictation accuracy, signature/addendum and viewer workflows.
- Load tests: worklist pagination, report search, concurrent viewer sessions and PACS timeout behavior.

13. Engineering implementation sequence

Sprint / phase	Deliverables
0: discovery	Architecture decision records, threat model, clinical scope, integration inventory
1-2: foundation	Repo, CI/CD, auth, tenant, DB migrations, logging, audit
3-4: operations	Patient registry, orders, scheduling, worklist and status engine
5-6: reporting	Templates, draft/versioning, signing, addenda, PDF export
7: imaging	Orthanc sandbox, DICOMweb adapter, OHIF launch, study linkage
8: AI assist	Dictation/LLM gateway, proposals, human review, prompt evaluation
9: readiness	Security testing, backups, clinical UAT, monitoring and pilot runbook

14. Regulatory and clinical safety gates

Before any clinical deployment: validate intended use and applicable Kosovo rules with local counsel and regulators; review medical-device classification; appoint clinical safety owner; perform data-protection impact assessment where required; establish vendor and patient-data contracts; validate clinical display and reporting processes. Later image-based diagnostic AI requires a distinct product/regulatory and evidence workstream.

15. Future interoperability

Use HL7 FHIR resources such as Patient, ServiceRequest, ImagingStudy, DiagnosticReport and Observation through a versioned adapter. Align integration contracts with actual IHIS requirements when published. Avoid hard-coded assumptions about national APIs.

References and evidence boundaries

World Bank, Kosovo healthcare strengthening (2024) — https://www.worldbank.org/en/news/press-release/2024/05/07/world-bank-supports-enhancement-of-health-services-in-kosovo-improving-the-lives-of-its-people

Kosovo Ministry of Health, IHIS oversight procurement (2026) — https://msh.rks-gov.net/Documents/Index/1?year=2026

World Bank, Kosovo Digital Health Blueprint case study — https://thedocs.worldbank.org/en/doc/b929829fc3d36bebdf9d93555c64f222-0140022024/related/Digital-Health-Blueprint-Toolkit-Kosovo-Case-Study.pdf

HL7 FHIR specification — https://hl7.org/fhir/

OHIF documentation — https://docs.ohif.org/

OHIF DICOMweb and Orthanc integration — https://docs.ohif.org/configuration/datasources/dicom-web/

Market demand, exact waiting times, clinic counts, current vendor penetration, willingness to pay, and legal classification require local primary research. All numerical commercial scenarios in this document are illustrative assumptions.